

Java Increment & Decrement

Advanced Expression-Based Questions

30 practice questions + 5 boss challenges. Output solve karne se pehle left-to-right dry run karo.

Level 1 — Pre/Post Increment & Decrement

Q1.

```
int a = 5;
int b = ++a + a++;
System.out.println(a + " " + b);
```

Output: _____

Q2.

```
int x = 10;
int y = x++ + ++x;
System.out.println(x + " " + y);
```

Output: _____

Q3.

```
int a = 7;
int b = a-- + --a;
System.out.println(a + " " + b);
```

Output: _____

Q4.

```
int x = 4;
int y = ++x + x++ + x;
System.out.println(x + " " + y);
```

Output: _____

Q5.

```
int n = 8;
int result = n-- + --n + n++;
System.out.println(n + " " + result);
```

Output: _____

Level 2 — Arithmetic Expressions

Q6.

```
int a = 3;
int b = 5;
int c = ++a * b++ + a;
System.out.println(a + " " + b + " " + c);
```

Output: _____

Q7.

```
int x = 6;
int y = 2;
int z = x++ * ++y + --x;
System.out.println(x + " " + y + " " + z);
```

Output: _____

Q8.

```
int a = 10;
int b = 20;
int result = a++ + --b + ++a;
System.out.println(a + " " + b + " " + result);
```

Output: _____

Q9.

```
int p = 2;
int q = 3;
int r = p++ + q++ * ++p;
System.out.println(p + " " + q + " " + r);
```

Output: _____

Q10.

```
int x = 5;
int result = x++ * x-- + ++x;
System.out.println(x + " " + result);
```

Output: _____

Level 3 — Multiple Variables & Assignment

Q11.

```
int a = 5;
int b = ++a;
int c = b++;
System.out.println(a + " " + b + " " + c);
```

Output: _____

Q12.

```
int x = 10;
int y = x++ + 5;
x = ++y;
System.out.println(x + " " + y);
```

Output: _____

Q13.

```
int a = 4;
int b = 6;
a = a++ + ++b;
System.out.println(a + " " + b);
```

Output: _____

Q14.

```
int x = 3;
int y = 7;
x += y++;
y = ++x;
System.out.println(x + " " + y);
```

Output: _____

Q15.

```
int p = 5;
int q = 10;
int r = p++ + q;
p = ++r;
System.out.println(p + " " + q + " " + r);
```

Output: _____

Level 4 — Conditions & Short-Circuit

Q16.

```
int x = 5;
if (x++ > 5 && ++x > 6) {
    System.out.println("YES");
}
System.out.println(x);
```

Output: _____

Q17.

```
int x = 5;
if (++x == 6 || x++ == 7) {
    System.out.println("TRUE");
}
System.out.println(x);
```

Output: _____

Q18.

```
int a = 3;
int b = 4;
if (a++ > 3 || ++b > 4) {
    System.out.println(a + " " + b);
}
```

Output: _____

Q19.

```
int n = 5;
if (n-- == 5 && --n == 3) {
    System.out.println("MATCH");
}
System.out.println(n);
```

Output: _____

Q20.

```
int x = 2;
if (x++ == 2 && ++x == 4 && x++ == 4) {
    System.out.println("PASS");
}
System.out.println(x);
```

Output: _____

Level 5 — Loops

Q21.

```
int i = 1;
while (i++ <= 5) {
    System.out.print(i + " ");
}
```

Output: _____

Q22.

```
int i = 5;
while (--i > 1) {
    System.out.print(i + " ");
}
```

Output: _____

Q23.

```
for (int i = 0; i++ < 5; ) {  
    System.out.print(i + " ");  
}
```

Output: _____

Q24.

```
for (int i = 5; i-- > 0; ) {  
    System.out.print(i + " ");  
}
```

Output: _____

Q25.

```
int i = 1;  
int sum = 0;  
while (i++ <= 5) {  
    sum += i;  
}  
System.out.println(sum);
```

Output: _____

Level 6 — Ternary

Q26.

```
int x = 4;  
int y = (x++ > 4) ? ++x : x--;  
System.out.println(x + " " + y);
```

Output: _____

Q27.

```
int a = 5;  
int b = ++a > 5 ? a++ : --a;  
System.out.println(a + " " + b);
```

Output: _____

Q28.

```
int n = 3;  
int result = (n++ == 3) ? n++ + ++n : --n;  
System.out.println(n + " " + result);
```

Output: _____

Q29.

```
int x = 10;  
int y = x-- > 10 ? ++x : x++ + ++x;  
System.out.println(x + " " + y);
```

Output: _____

Q30.

```
int p = 2;  
int q = (p++ > 2) ? ++p : p++ + 5;  
System.out.println(p + " " + q);
```

Output: _____

Boss Level — Challenge

Q31.

```
int x = 5;
int y = x++ + ++x + x-- + --x;
System.out.println(x + " " + y);
```

Output: _____

Q32.

```
int a = 2;
int b = 3;
int c = a++ + ++b * a++ - --b;
System.out.println(a + " " + b + " " + c);
```

Output: _____

Q33.

```
int x = 1;
int y = 2;
int z = ++x + y++ + x++ + ++y;
System.out.println(x + " " + y + " " + z);
```

Output: _____

Q34.

```
int n = 5;
int result = n++ + (n > 5 ? ++n : n--) + --n;
System.out.println(n + " " + result);
```

Output: _____

Q35.

```
int x = 3;
if (x++ == 3 && x++ == 4 || ++x == 6) {
    System.out.println("JAVA");
}
System.out.println(x);
```

Output: _____

Trainer Notes

- **Pre-increment ($++x$):** pehle value increase, phir expression me use.
- **Post-increment ($x++$):** pehle current value use, phir increase.
- **Pre/Post decrement:** same rule, lekin value 1 se decrease hoti hai.
- **Java evaluation:** operands ko left-to-right track karo.
- **Short-circuit:** $\&\&$ me left false ho to right side skip; $\|\|$ me left true ho to right side skip.
- **Classroom tip:** har operand ke baad variable ki current value board par likho.
- **Production tip:** repeated side effects wale expressions puzzles ke liye useful hain, lekin real code me readability ke liye avoid karna better hai.